What Robot Data Should We Collect Next?

A Cost-Aware Study of Recovery-and-Correction Post-Training for Low-Cost Bimanual Manipulation

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Draft status. This document is an experimental blueprint. All orange values and every result in Sec. VII are synthetic examples chosen only to illustrate the intended analysis. They are not experimental observations and must be replaced before public release.

***Abstract*—Improving a robot policy requires choosing not only how much data to collect, but what kind of data to collect next. Real-robot behavior cloning (BC), simulation, egocentric human demonstrations, and human-in-the-loop recovery data impose different costs and may be valuable at different stages of policy development. We propose a cost-aware study of these alternatives on a low-cost, fixed-base, 12-degree-of-freedom bimanual robot. Starting from the same multi-task $\pi_{0.5}$ checkpoint trained on target-robot BC, simulation, and aligned egocentric data, we add matched increments of (i) clean BC, (ii) simulated demonstrations, (iii) egocentric demonstrations, (iv) correction-only interventions, or (v) Recovery and Correction (RaC) interventions. We measure macro-averaged complete success over eight tasks, task-level success, controlled-perturbation recovery, and retention of nominal skills. Effort is reported as a vector of active human time, robot occupancy, engineering time, compute, and elapsed turnaround; a scalar cost is used only in explicitly stated deployment scenarios. Source-specific scaling curves estimate both average and marginal success gain per unit effort, while Pareto frontiers identify non-dominated data-collection strategies. The primary test compares RaC with additional clean demonstrations and correction-only data at equal human effort and fixed optimization budget. Illustrative synthetic results show how the proposed analysis could reveal a late-stage advantage for RaC on long-horizon tasks while simulation remains attractive when physical robot time is scarce. The study is intended to produce decision guidance for resource-constrained robot-learning projects, not a universal ranking of data sources.**

***Index Terms*—robot learning, bimanual manipulation, vision-language-action models, recovery, human intervention, data scaling, cost-aware data collection**

I. Introduction

Robot-learning projects face a recurring allocation decision: after a policy has reached moderate performance, should the next unit of effort be spent on more successful target-robot demonstrations, more simulated trajectories, more human video, or examples of the policy's own failures and recoveries? These sources are not interchangeable. Clean behavior-cloning data align the policy with the target embodiment but require scarce operator and robot time. Simulation can generate large trajectory volumes without occupying the physical robot, but requires an environment, controllers, and sim-to-real engineering. Egocentric human data can broaden object and scene coverage but introduces a cross-embodiment supervision gap. Recovery data target states that a deployed policy actually visits, but can only be collected after a usable policy exists.

Standard ablation studies typically compare one final model per data source. Such comparisons establish whether a source helped at one chosen scale, but do not answer three practical questions: how quickly its benefit saturates, when another source becomes preferable, and whether its apparent efficiency survives the inclusion of fixed setup costs. A useful allocation study therefore needs multiple data-budget levels, a common starting checkpoint, matched training budgets, and explicit accounting of resources that cannot be converted into one another without assumptions.

This report focuses on Recovery and Correction (RaC), an iterative human-in-the-loop protocol introduced to teach retry and adaptation behavior in long-horizon imitation policies [7]. When a failure appears imminent, the operator first returns the robot toward a familiar state and then demonstrates a correction that completes the current subtask. RaC differs from correction-only intervention because part of the demonstration deliberately moves backward in task progress. The original study found that this data composition could outperform clean demonstrations and correction-only interventions, and explicitly identified application to generalist vision-language-action (VLA) policies as future work.

We study RaC on the same low-cost bimanual platform and task suite used to train a heterogeneous $\pi_{0.5}$ baseline. The goal is not to claim a new recovery algorithm. Instead, we ask a resource-allocation question:

> For a multi-task VLA that has already learned from robot BC, simulation, and egocentric data, what is the incremental return of RaC relative to spending the same effort on additional data from those sources?

The phrase *return* always denotes measured policy improvement; the phrase *effort* denotes a reported resource vector. Monetary return on investment is not claimed unless an explicit price model is supplied.

a) Contributions.: This study is designed to make four contributions:

- A matched-start, matched-optimization comparison of clean

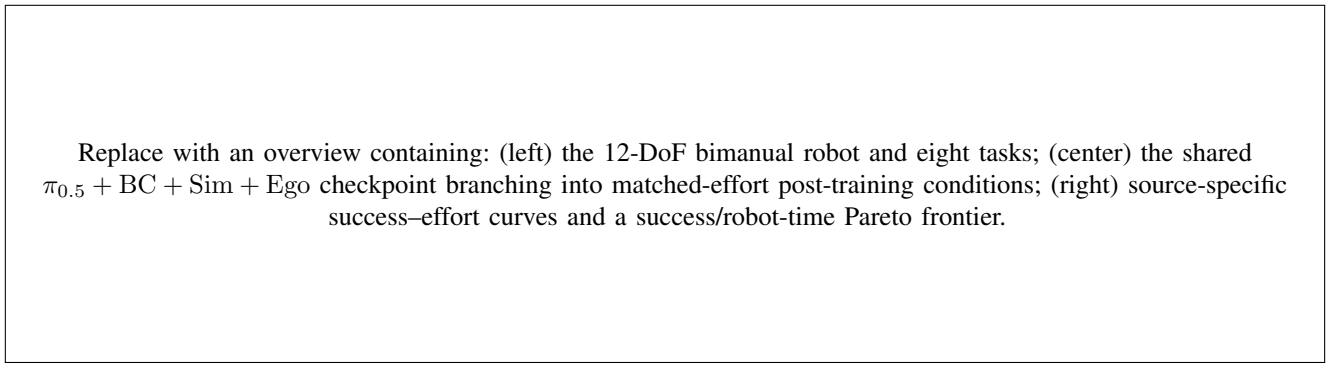


Fig. 1. Study overview. Every branch starts from the same checkpoint, uses the same number of optimizer updates and original-data replay schedule, and differs only in the newly acquired data. RaC interventions contain a recovery segment followed by a correction segment.

BC, simulation, aligned egocentric data, correction-only intervention, and RaC post-training on one bimanual VLA.
- Source-specific learning curves that estimate average and incremental percentage-point gains per active collection hour, together with uncertainty and per-task results.
- A multi-resource accounting protocol that separates marginal collection effort from fully loaded pipeline cost and reports Pareto-optimal strategies rather than hiding value judgments inside one cost number.
- A direct test of whether RaC improves controlled recovery and long-horizon completion without degrading nominal performance on the original eight-task suite.

II. Related Work

a) Imitation learning and intervention.: Behavior cloning trains a policy on expert state–action pairs but can fail when its own errors move execution outside the demonstration distribution. DAgger addresses this mismatch by aggregating labels on states visited by the learned policy [1]. HG-DAgger adapts interactive imitation to human interventions [2]. These methods motivate collecting data from policy-induced states rather than only increasing a static expert dataset.

b) Recovery and correction.: RaC structures each intervention as recovery toward a familiar state followed by correction of the current subtask, and terminates the episode after the intervention [7]. Its learning objective remains imitation; the change is in trajectory composition. Recent work also studies failure synthesis, reasoning, and recovery for VLA policies [9], [10], making it important to state that the present contribution is a cost-controlled evaluation of a known protocol, not the first VLA recovery system.

c) Heterogeneous VLA training data.: The original π_0 and its successor $\pi_{0.5}$ combine vision-language pretraining with continuous action generation and heterogeneous robot data [3], [4]. Human-to-robot transfer work suggests that aligned egocentric experience can add useful manipulation supervision to a VLA [5], while EgoScale studies improvement with large amounts of action-labeled egocentric video [6]. These results motivate treating human video as one candidate investment rather than assuming that every additional hour has the same value as robot interaction.

d) Scaling curves and data allocation.: Data-scaling studies characterize how performance changes with dataset size. Factored Scaling Curves use such relationships to prioritize demonstrations for different environment factors under a fixed collection budget [8]. Our study is complementary: its axes are acquisition sources with different fixed costs, action labels, embodiments, and ordering constraints. In particular, RaC is endogenous to the current policy because the policy determines which failures are observed.

III. Robot Platform and Task Suite

The platform is the same custom, low-cost, fixed-base bimanual robot used in the preceding benchmark study. It contains two 6-DoF arms for 12 controlled arm degrees of freedom, two grippers, two fixed RGB cameras, and one RGB wrist camera per arm. The controller runs at $50\,\mathrm{Hz}$. Policies observe the four RGB views, proprioception, and a language instruction. $\pi_{0.5}$ predicts relative dual-arm Cartesian end-effector trajectories and gripper commands; inverse kinematics maps these targets to joint commands.

The eight tasks, in approximate order of difficulty, are:

T1. place a cube inside a target container;
T2. retrieve a cube from a container;
T3. sort blue and green cubes by color;
T4. open a language-specified drawer;
T5. stack one cube on another;
T6. perform a bimanual mid-air handover and placement;
T7. stock a requested cube in a specified drawer and close it;
T8. retrieve a cube from a specified drawer and close it.

Tasks T1–T6 last approximately $10\,\mathrm{s}$ under nominal execution; T7–T8 last approximately $20\,\mathrm{s}$. We prespecify T1–T6 as the short/medium-horizon group and T7–T8 as the long-horizon group. This grouping is reported in addition to, not instead of, individual task results.

IV. Problem Formulation

A. Performance estimands

Let $Y_{t,r}(\pi) \in \{0, 1\}$ indicate complete success of policy π on task t and rollout r. The task success probability is

$$p_t(\pi) = \mathbb{E}[Y_{t,r}(\pi)]. \tag{1}$$

The primary aggregate is the equally weighted macro success rate

$$S_{\text{macro}}(\pi) = \frac{1}{8} \sum_{t=1}^{8} p_t(\pi). \tag{2}$$

Equal weighting prevents tasks with more or shorter rollouts from dominating the estimate. We additionally report every p_t, the T1–T6 and T7–T8 group means, and mean subtask progress. Aggregation is never used to conceal a task-level regression.

For standardized perturbation q, let $R_{t,q,r}(\pi)$ indicate completion of the original instruction after the perturbation without human assistance. The recovery rate S_{rec} is reported separately from nominal success. We do not average nominal and recovery outcomes into the primary metric.

B. Effort accounting

For source j, effort is the vector

$$\mathbf{c}_j = (H_j, R_j, E_j, G_j, W_j), \tag{3}$$

where H is active human collection or annotation time, R is physical robot occupancy, E is engineering/setup time, G is accelerator time, and W is elapsed wall-clock turnaround. Data duration and number of transitions are reported alongside this vector but are not treated as costs by themselves.

We distinguish two views:

- **Marginal operating effort** excludes already completed pipeline development and answers what should be collected next on the existing system.
- **Fully loaded effort** includes source-specific setup and answers whether building that capability was worthwhile over a stated number of tasks or projects.

If a scalar cost is needed for a deployment scenario s, it is computed as

$$C_j^{(s)} = \boldsymbol{\lambda}^{(s)\top} \mathbf{c}_j, \tag{4}$$

where all weights $\boldsymbol{\lambda}^{(s)}$ are published. Conclusions are tested across multiple plausible weight settings.

C. Average and marginal return

Let $\pi_j(h)$ be the policy after adding source j using h active human hours, and let π_0 be the shared baseline. Average incremental return is

$$\text{AIR}_j(h) = \frac{S_{\text{macro}}(\pi_j(h)) - S_{\text{macro}}(\pi_0)}{h}. \tag{5}$$

Because gains typically diminish, the decision-relevant quantity between two budgets is

$$\text{MIR}_j(h_a, h_b) = \frac{S_{\text{macro}}(\pi_j(h_b)) - S_{\text{macro}}(\pi_j(h_a))}{h_b - h_a}. \tag{6}$$

Both quantities are percentage points per active hour, not universal economic returns. We also report the fraction of baseline failure headroom removed,

$$\text{HGR}_j(h) = \frac{S_{\text{macro}}(\pi_j(h)) - S_{\text{macro}}(\pi_0)}{1 - S_{\text{macro}}(\pi_0)}. \tag{7}$$

D. Pareto optimality

A policy is dominated if another policy achieves at least as much success with no more of every reported resource and is strictly better on at least one dimension. The remaining policies form the Pareto frontier. We report separate two-dimensional frontiers for success versus human time, robot time, and turnaround, plus the non-dominated set under the full effort vector. This avoids claiming a unique best source when laboratories face different constraints.

V. Data Sources and Post-Training Conditions

A. Shared starting checkpoint

All branches begin from the same eight-task $\pi_{0.5}$ checkpoint trained on target-robot BC, MuJoCo simulation, and aligned RGB-D egocentric human demonstrations. We denote it $\pi_{0.5}^{\text{BSE}}$, where B, S, and E denote BC, simulation, and ego data. The checkpoint, optimizer state, normalization statistics, task sampling, and random seeds are frozen before new data collection.

B. Candidate investments

a) Continued training.: No data are added. The baseline dataset is sampled for the same number of optimizer updates as the data-addition branches. This controls for additional optimization.

b) Additional BC.: The operator collects complete successful trajectories using the existing leader–follower interface. Initial conditions are sampled from the same distribution as the test suite. Failed demonstrations are discarded according to the original BC protocol, but their collection time remains charged.

c) Additional simulation.: The existing MuJoCo environments generate successful demonstrations under the same domain-randomization family as the baseline dataset. Human time spent authoring new task logic or randomization is charged as engineering effort; automated rollout and rendering time are recorded separately.

d) Additional egocentric data.: The operator performs the tasks while wearing the RGB-D camera. Hand poses are converted into the same relative bimanual end-effector representation used in the baseline study. Recording, calibration, automatic processing, and manual quality-control time are logged separately. Human gripper labels follow the predefined proxy used by the baseline and are not described as ground-truth robot actions.

e) Correction-only intervention.: The current policy is executed until an operator judges that failure is imminent. The operator takes over and demonstrates actions that directly advance or complete the current subtask. This is the principal human-in-the-loop control for determining whether recovery segments add value beyond corrections.

f) RaC intervention.: RaC follows two prespecified rules [7]:

1) On takeover, the operator first executes a recovery segment that returns the robot or manipulated object toward a familiar state, and then executes a correction that completes the current subtask.
2) The rollout terminates after the intervention concludes.

Only the human-controlled transitions are retained from an intervened rollout. An autonomous rollout is retained only if it completes the entire task without intervention. Each transition stores policy/human control status and recovery/correction phase labels. A second annotator audits a prespecified sample of phase boundaries.

C. Matched training

For every condition and effort level, the number of optimizer updates, batch size, learning-rate schedule, augmentation, and checkpoint-selection rule are identical. Original BSE data occupy a fixed 75% of training batches and the new source occupies 25%; these fractions are illustrative hyperparameters to be selected before viewing physical test results. The same original-data replay rate reduces catastrophic forgetting and prevents a large new dataset from silently receiving more optimization. Continued training uses only original BSE data for all batches.

VI. Experimental Protocol

A. Effort levels and collection order

New data are collected cumulatively at $h \in \{1, 2, 4\}$ active human-hour budgets across the eight tasks. Within each source and round, effort is divided equally across tasks; unused time caused by early completion is reassigned by a published deterministic rule. The source order is randomized across collection days to reduce temporal confounding from lighting, hardware wear, and operator learning. Every failure, rejected trajectory, reset, and unusable recording is included in the effort ledger.

TABLE I
Post-training conditions. All data branches start from the same $\pi_{0.5}^{\mathrm{BSE}}$ checkpoint.

Condition	New data	On-policy	Recovery
Continue	None	No	No
+BC	Full successes	No	Incidental
+Sim	Sim successes	No	Scripted only
+Ego	Human task demos	No	Incidental
+Correction	Corrections	Yes	No
+RaC	Recovery + correction	Yes	Explicit

RaC and correction-only data are collected iteratively. After each effort round, the corresponding policy is retrained and becomes the rollout policy for the next round. BC, simulation, and ego branches also use cumulative datasets, but their collection does not depend on the branch policy. This difference is part of the intervention being evaluated and is stated explicitly.

B. Evaluation

Each checkpoint is evaluated on matched blocks of initial conditions. A block contains one rollout from every condition in randomized order, with object placements and language instructions held constant to the extent physically possible. Evaluators are blinded to policy condition when scoring videos.

For the dense effort curves, the protocol uses 30 physical rollouts per task and checkpoint. The baseline, the best non-RaC comparator selected by the preregistered aggregate metric, and the final RaC checkpoint are additionally trained from three independent seeds and evaluated for 20 rollouts per task and seed. Simulation uses at least three seeds at every budget. These counts are proposed values and must be confirmed by a prospective power analysis using pilot variability.

a) Nominal evaluation.: The primary outcome is complete task success. Partial progress is the number of prespecified subtasks completed before irrecoverable failure. Maximum execution time and retry count are fixed across policies so that RaC cannot obtain an unbounded advantage by acting indefinitely.

b) Controlled recovery evaluation.: Three event-triggered perturbations are applied where relevant: target-object displacement before grasp, replacement of a lifted object at a nearby reachable pose, and partial drawer reset after opening. Perturbation magnitudes and trigger events are frozen before evaluation. A fourth held-out perturbation family is excluded from all recovery collection and used to test generalization.

c) Retention.: All eight nominal tasks are retested after each post-training condition. A source is not considered beneficial if

aggregate improvement is driven by long-horizon tasks while previously reliable skills regress beyond the predeclared non-inferiority margin of 5 percentage points.

C. Statistical analysis

We report raw counts, task-level Wilson intervals, and a task-stratified bootstrap interval for macro success. The primary contrast is final-budget RaC minus the best matched-effort non-RaC condition. Secondary contrasts compare RaC with correction-only and additional BC. Training-seed variability is reported separately from rollout variability.

Effort curves are fitted with monotone saturating splines; a log-linear or power law is reported only if residual checks support it. Uncertainty in AIR and MIR is obtained by applying the complete calculation inside each bootstrap draw. Pareto membership is reported both for point estimates and as the fraction of bootstrap draws in which a policy is non-dominated. No claim of universal source ordering is made from overlapping intervals.

The following analysis decisions are frozen before physical evaluation: task weights, success criteria, excluded-rollout rules, perturbation triggers, non-inferiority margin, budget levels, source cost definitions, and the rule for choosing the strongest non-RaC comparator.

VII. Illustrative Expected Results

Synthetic example only. Every number in this section is invented to show the intended tables, calculations, and possible interpretation. No claim about the robot or policy may be based on these values.

A. Aggregate benefit as effort increases

Table II illustrates the central result. In this mock scenario, all additional data sources improve on the baseline, but RaC has the steepest late-stage curve. Correction-only data outperform additional clean BC, suggesting value from visiting policy-induced states, while the further RaC gain suggests that recovery segments contribute beyond corrections.

The corresponding illustrative headroom-gain ratio for RaC is $(0.795 - 0.594)/(1 - 0.594) = 49.5\%$: approximately half of the baseline's remaining failures are removed. This is more informative than the absolute 20.1-point gain alone, but it remains conditional on this task suite, baseline, and budget.

The example also illustrates why average ROI should not be extrapolated. RaC's mock marginal returns are 6.6, 6.1, and 3.7 percentage points per hour over the 0–1, 1–2, and 2–4 h intervals, respectively. The decrease indicates saturation; it does not justify predicting another 20.1-point improvement from the next four hours.

B. Where the gain occurs

The synthetic per-task results in Table III place most of RaC's advantage on the handover and two warehouse tasks. This would be consistent with the hypothesis that retry behavior is most valuable when errors compound, but the causal interpretation would require the correction-only comparison and annotated recovery events. Smaller gains on T1–T2 would establish a useful boundary rather than a failure of the method.

C. Recovery and nominal-skill retention

Table IV shows an illustrative recovery result. The key comparison is not only whether RaC has the highest perturbed success, but whether it retains clean success and generalizes to a perturbation excluded from its training data. A large gain only on intervention-matched failures would support a narrower claim than general recovery.

In this mock outcome, no original task regresses relative to the baseline, so the retention criterion is satisfied. If measured results instead reveal, for example, improved warehouse recovery but degraded sorting, the paper must report that trade-off and test whether stronger replay resolves it.

D. Effort vectors and Pareto interpretation

Equal active hours do not imply equal resource use. Table V illustrates a transparent ledger for one 4-h increment. Simulation and ego data would remain attractive for laboratories with little robot access even if their aggregate success gains were smaller. Conversely, RaC could be attractive when the platform already exists and operator time, rather than robot occupancy, is the binding constraint.

Under the synthetic values, RaC dominates the other strategies on the success-versus-active-human-time plot because every source receives four active hours. It does *not* dominate simulation or ego data under the full effort vector, because those sources require no physical robot occupancy. The non-dominated set therefore depends on the laboratory's constraint, which is exactly why the Pareto frontier is reported.

For fully loaded comparisons, fixed setup is amortized over N tasks or future projects:

$$\bar{C}_j(N) = \frac{F_j}{N} + V_j, \tag{8}$$

where F_j is fixed setup effort and V_j is the marginal effort for the evaluated increment. Break-even values of N are reported with sensitivity to the cost weights in Eq. (4); they are not inferred from the illustrative ledger above.

TABLE II

SYNTHETIC ILLUSTRATION—NOT MEASURED RESULTS. MACRO COMPLETE SUCCESS (%) OVER EIGHT EQUALLY WEIGHTED TASKS AS A FUNCTION OF ACTIVE COLLECTION EFFORT. INTERVALS SHOWN ONLY AT THE FINAL BUDGET ARE ILLUSTRATIVE TASK-STRATIFIED 95% BOOTSTRAP INTERVALS.

Condition	0 h	1 h	2 h	4 h	4-h gain / AIR
Continue	59.4	59.5	59.8	59.8 [55.5, 64.0]	+0.4 / +0.1 pp/h
+Ego	59.4	60.8	62.5	64.1 [59.8, 68.2]	+4.7 / +1.2 pp/h
+Simulation	59.4	61.9	64.5	67.3 [63.0, 71.3]	+7.9 / +2.0 pp/h
+BC	59.4	63.2	66.8	70.1 [66.0, 74.0]	+10.7 / +2.7 pp/h
+Correction	59.4	64.0	68.4	72.3 [68.2, 76.0]	+12.9 / +3.2 pp/h
+RaC	59.4	66.0	72.1	79.5 [75.8, 82.8]	+20.1 / +5.0 pp/h

Replace with measured curves. Panel A: macro nominal success versus active collection time for every source, with bootstrap bands. Panel B: recovery success versus effort. Panel C: marginal return between successive budgets. Panel D: fraction of baseline failure headroom removed.

Fig. 2. Source-specific effort curves. The final paper must plot individual measurements and uncertainty, not only fitted curves.

TABLE III

SYNTHETIC ILLUSTRATION—NOT MEASURED RESULTS. COMPLETE SUCCESS (%) AFTER 4 H OF ADDITIONAL ACTIVE COLLECTION EFFORT.

Condition	T1	T2	T3	T4	T5	T6	T7	T8	Macro
Baseline	85	82	70	68	58	50	32	30	59.4
+Ego	90	88	76	73	62	56	36	32	64.1
+Simulation	90	87	77	74	65	59	44	42	67.3
+BC	92	90	80	78	69	62	46	44	70.1
+Correction	91	89	81	80	71	66	52	48	72.3
+RaC	93	92	85	86	80	76	64	60	79.5

TABLE IV

SYNTHETIC ILLUSTRATION. AGGREGATE COMPLETION AFTER EVENT-TRIGGERED PERTURBATION (%). "HELD-OUT" DENOTES A PERTURBATION FAMILY NOT USED DURING INTERVENTION COLLECTION.

Policy	Nom.	Object	Drawer	Held-out
Baseline	59	31	28	24
+BC	70	39	36	31
+Correction	72	49	45	38
+RaC	80	67	63	53

VIII. DISCUSSION

A. What the proposed result would establish

If measured results follow the illustrative pattern, the justified conclusion would be narrow: after heterogeneous BSE training on this robot, RaC provides a larger local gain per active collection hour than adding more data from the compared sources, particularly on the tested long-horizon tasks. The result would not establish that RaC should be collected first, that it is cheapest for every laboratory, or that its advantage persists at much larger scales.

The comparison with correction-only data carries the main mechanistic evidence. If RaC and correction-only perform similarly, the study would show that on-policy intervention is useful but would not support a special benefit from deliberate recovery. If RaC improves only the failures seen during collection, the result would support targeted robustness rather than general recovery. Null and negative results remain informative for allocation.

B. When each source may be preferred

The intended output is a stage-conditional decision map, not one global winner. BC may have high early value when target-embodiment alignment is poor. Simulation may be preferred after its fixed environment cost is amortized or when robot access is scarce. Ego data may be most useful when visual and semantic coverage, rather than low-level control, is the bottleneck. RaC is inherently a later-stage investment: it

TABLE V

SYNTHETIC ILLUSTRATION—REPLACE WITH MEASURED LOGS. EFFORT FOR A NOMINAL 4-H ACTIVE COLLECTION INCREMENT. FIXED SETUP IS INCLUDED ONLY IN THE FULLY LOADED ANALYSIS. GPU HOURS INCLUDE DATA PROCESSING BUT EXCLUDE THE MATCHED POLICY-TRAINING BUDGET.

Source	Active human h	Robot h	GPU h	Wall-clock h	Fixed setup h
BC	4.0	5.0	0.5	6.0	6
Simulation	4.0	0.0	40.0	44.0	80
Ego	4.0	0.0	6.0	10.0	24
Correction	4.0	7.0	0.5	8.0	8
RaC	4.0	8.0	0.5	9.0	12

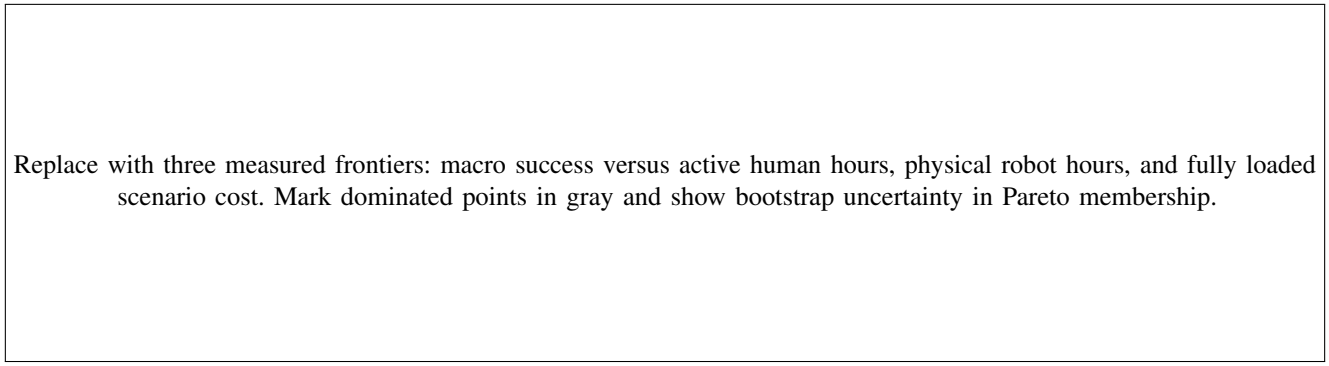


Fig. 3. Pareto analysis. A source may be dominated under one resource definition but remain non-dominated when another scarce resource is considered.

requires a policy capable enough to reach informative failure states and an operator able to recognize recoverable deviations.

C. Prospective allocation test

A stronger follow-up uses the first two budget points to predict which source or mixture will maximize success under a new held-out budget. The recommended allocation is then collected prospectively and compared with equal allocation, all-BC, and all-RaC strategies. This test distinguishes a useful allocation model from a retrospective description of already observed curves.

IX. LIMITATIONS

All experiments use one robot embodiment, one workspace, one operator interface, and eight tasks. Costs are institution-specific; active human time may be scarce in one laboratory while accelerator time is scarce in another. The same data source may have a different return with another VLA, action representation, or baseline competence level.

Data sources are not perfectly exchangeable. RaC and correction data are generated by the current policy, whereas BC, simulation, and ego data can be collected independently. The iterative RaC branch therefore benefits from targeting its own evolving errors. This is a defining property of the method, but it prevents interpreting the comparison as an architecture-free causal effect of individual frames.

The macro average assigns equal value to all tasks. Other deployments may value warehouse tasks more heavily; per-task results and alternative prespecified weights are therefore essential. Trial-level confidence intervals do not replace training-seed replication. A result from one baseline mixture cannot identify all interactions among BC, simulation, ego, and recovery data.

Finally, manual intervention judgments can vary across operators. The study should log takeover timing, intervention duration, phase labels, and inter-rater audit results. Safety stops, hardware faults, camera dropouts, and operator mistakes must be reported rather than silently removed.

X. CONCLUSION

This report proposes a cost-aware evaluation of what robot data to collect after a heterogeneous multi-task VLA has reached moderate competence. By comparing additional BC, simulation, egocentric demonstrations, correction-only interventions, and RaC from the same checkpoint and under matched optimization, the study separates additional training from additional data and recovery-aware data composition. Multiple effort levels expose diminishing returns; task-level and perturbation evaluations reveal where gains occur; and Pareto frontiers keep resource trade-offs visible. The illustrative numbers in this draft are not results. Once replaced by preregistered physical experiments, the analysis can support a concrete, bounded conclusion about when recovery-and-correction data is the best next investment for this robot and benchmark.

REPRODUCIBILITY AND REPORTING CHECKLIST

The final release should include:

1) hashes of the shared starting checkpoint, code, and frozen datasets;
2) raw per-rollout outcomes, task seeds, policy seeds, and exclusion logs;
3) source-specific accepted and rejected data with collection timestamps;
4) active human, robot occupancy, engineering, compute, and turnaround logs;
5) recovery/correction phase annotations and audit agreement;
6) exact replay mixture, optimizer updates, and checkpoint-selection rule;
7) perturbation definitions, magnitudes, triggers, and held-out family;
8) scripts that reconstruct every aggregate, interval, curve, and frontier;
9) videos of representative nominal successes, failures, recoveries, and regressions, including uncut evaluation blocks;
10) a machine-readable marker that identifies the current document's mock results until they have all been replaced.

REFERENCES

[1] S. Ross, G. Gordon, and J. A. Bagnell, "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning," in *Proc. AISTATS*, 2011.

[2] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer, "HG-DAgger: Interactive Imitation Learning with Human Experts," in *Proc. IEEE International Conference on Robotics and Automation*, 2019.

[3] K. Black et al., "π_0: A Vision-Language-Action Flow Model for General Robot Control," *arXiv preprint arXiv:2410.24164*, 2024.

[4] K. Black et al., "$\pi_{0.5}$: A Vision-Language-Action Model with Open-World Generalization," *arXiv preprint arXiv:2504.16054*, 2025.

[5] S. Kareer, K. Pertsch, J. Darpinian, J. Hoffman, D. Xu, S. Levine, C. Finn, and S. Nair, "Emergence of Human to Robot Transfer in Vision-Language-Action Models," *arXiv preprint arXiv:2512.22414*, 2025.

[6] R. Zheng et al., "EgoScale: Scaling Dexterous Manipulation with Diverse Egocentric Human Data," *arXiv preprint arXiv:2602.16710*, 2026.

[7] Z. Hu, R. Wu, N. Enock, J. Li, R. Kadakia, Z. Erickson, and A. Kumar, "RaC: Robot Learning for Long-Horizon Tasks by Scaling Recovery and Correction," *arXiv preprint arXiv:2509.07953*, 2025.

[8] L. Zha, A. Badithela, M. Zhang, J. Lidard, J. Bao, E. Zhou, D. Snyder, A. Z. Ren, D. Shah, and A. Majumdar, "Guiding Data Collection via Factored Scaling Curves," *arXiv preprint arXiv:2505.07728*, 2025.

[9] Z. Lin, J. Duan, H. Fang, D. Fox, R. Krishna, C. Tan, and B. Wen, "FailSafe: Reasoning and Recovery from Failures in Vision-Language-Action Models," *arXiv preprint arXiv:2510.01642*, 2025.

[10] W. Liufu et al., "RePO-VLA: Recovery-Driven Policy Optimization for Vision-Language-Action Models," *arXiv preprint arXiv:2605.09410*, 2026.